



June 27, 2024

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Effluent PFAS
Pace Project No.: 50374521

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on May 30, 2024. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in blue ink that reads 'Brian Hall'.

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Effluent PFAS

Pace Project No.: 50374521

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414

Alabama Certification #: 40770

Alaska Contaminated Sites Certification #: 17-009

Alaska DW Certification #: MN00064

Arizona Certification #: AZ0014

Arkansas DW Certification #: MN00064

Arkansas WW Certification #: 88-0680

California Certification #: 2929

Colorado Certification #: MN00064

Connecticut Certification #: PH-0256

DoD Certification via A2LA #: 2926.01

EPA Region 8 Tribal Water Systems+Wyoming DW
Certification #: via MN 027-053-137

Florida Certification #: E87605

Georgia Certification #: 959

GMP+ Certification #: GMP050884

Hawaii Certification #: MN00064

Idaho Certification #: MN00064

Illinois Certification #: 200011

Indiana Certification #: C-MN-01

Iowa Certification #: 368

ISO/IEC 17025 Certification via A2LA #: 2926.01

Kansas Certification #: E-10167

Kentucky DW Certification #: 90062

Kentucky WW Certification #: 90062

Louisiana DEQ Certification #: AI-03086

Louisiana DW Certification #: MN00064

Maine Certification #: MN00064

Maryland Certification #: 322

Michigan Certification #: 9909

Minnesota Certification #: 027-053-137

Minnesota Dept of Ag Approval: via MN 027-053-137

Minnesota Petrofund Registration #: 1240

Mississippi Certification #: MN00064

Missouri Certification #: 10100

Montana Certification #: CERT0092

Nebraska Certification #: NE-OS-18-06

Nevada Certification #: MN00064

New Hampshire Certification #: 2081

New Jersey Certification #: MN002

New York Certification #: 11647

North Carolina DW Certification #: 27700

North Carolina WW Certification #: 530

North Dakota Certification (A2LA) #: R-036

North Dakota Certification (MN) #: R-036

Ohio DW Certification #: 41244

Ohio VAP Certification (1700) #: CL101

Oklahoma Certification #: 9507

Oregon Primary Certification #: MN300001

Oregon Secondary Certification #: MN200001

Pennsylvania Certification #: 68-00563

Puerto Rico Certification #: MN00064

South Carolina Certification #: 74003001

Tennessee Certification #: TN02818

Texas Certification #: T104704192

Utah Certification #: MN00064

Vermont Certification #: VT-027053137

Virginia Certification #: 460163

Washington Certification #: C486

West Virginia DEP Certification #: 382

West Virginia DW Certification #: 9952 C

Wisconsin Certification #: 999407970

Wyoming UST Certification via A2LA #: 2926.01

USDA Permit #: P330-19-00208

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SAMPLE SUMMARY

Project: Effluent PFAS

Pace Project No.: 50374521

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50374521001	Effluent Duplicate	Water	05/29/24 13:27	05/30/24 10:15
50374521002	Effluent	Water	05/29/24 13:25	05/30/24 10:15
50374521003	Field Blank	Water	05/29/24 13:23	05/30/24 10:15

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SAMPLE ANALYTE COUNT

Project: Effluent PFAS

Pace Project No.: 50374521

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50374521001	Effluent Duplicate	ENV-SOP-MIN4-0178	MJL	53	PASI-M
50374521002	Effluent	ENV-SOP-MIN4-0178	MJL	53	PASI-M
50374521003	Field Blank	ENV-SOP-MIN4-0178	MJL	53	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

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SUMMARY OF DETECTION

Project: Effluent PFAS

Pace Project No.: 50374521

Lab Sample ID	Client Sample ID					
Method	Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50374521001	Effluent Duplicate					
ENV-SOP-MIN4-0178	ADONA	0.46J	ng/L	1.9	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFBS	25.9	ng/L	1.8	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFDA	0.34J	ng/L	2.1	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFHxA	7.6	ng/L	2.1	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFBA	6.8	ng/L	2.1	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFPeA	2.2	ng/L	2.1	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFHpA	0.51J	ng/L	2.1	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFHxS	0.35J	ng/L	1.9	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFNA	0.27J	ng/L	2.1	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFOS	1.1J	ng/L	1.9	06/27/24 13:57	
ENV-SOP-MIN4-0178	PFOA	1.7J	ng/L	2.1	06/27/24 13:57	
50374521002	Effluent					
ENV-SOP-MIN4-0178	ADONA	0.46J	ng/L	1.9	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFBS	25.3	ng/L	1.7	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFDA	0.35J	ng/L	2.0	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFHxA	7.4	ng/L	2.0	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFBA	7.5	ng/L	2.0	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFPeA	1.5J	ng/L	2.0	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFHpA	0.65J	ng/L	2.0	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFHxS	0.66J	ng/L	1.8	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFOS	0.99J	ng/L	1.8	06/27/24 13:50	
ENV-SOP-MIN4-0178	PFOA	1.7J	ng/L	2.0	06/27/24 13:50	

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50374521

Sample: Effluent Duplicate Lab ID: 50374521001 Collected: 05/29/24 13:27 Received: 05/30/24 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
PFAS ID NPW									
Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178									
Pace Analytical Services - Minneapolis									
11CI-PF3OUdS	<0.42	ng/L	1.9	0.42	1	06/25/24 11:00	06/27/24 13:57	763051-92-9	
4:2 FTS	<0.37	ng/L	1.9	0.37	1	06/25/24 11:00	06/27/24 13:57	757124-72-4	
6:2 FTS	<0.59	ng/L	2.0	0.59	1	06/25/24 11:00	06/27/24 13:57	27619-97-2	
8:2 FTS	<0.81	ng/L	2.0	0.81	1	06/25/24 11:00	06/27/24 13:57	39108-34-4	
9CI-PF3ONS	<0.36	ng/L	1.9	0.36	1	06/25/24 11:00	06/27/24 13:57	756426-58-1	
ADONA	0.46J	ng/L	1.9	0.33	1	06/25/24 11:00	06/27/24 13:57	919005-14-4	
HFPO-DA	<0.26	ng/L	2.1	0.26	1	06/25/24 11:00	06/27/24 13:57	13252-13-6	
NEtFOSAA	<0.58	ng/L	2.1	0.58	1	06/25/24 11:00	06/27/24 13:57	2991-50-6	
NMeFOSAA	<0.80	ng/L	2.1	0.80	1	06/25/24 11:00	06/27/24 13:57	2355-31-9	
PFBS	25.9	ng/L	1.8	0.21	1	06/25/24 11:00	06/27/24 13:57	375-73-5	
PFDA	0.34J	ng/L	2.1	0.26	1	06/25/24 11:00	06/27/24 13:57	335-76-2	
PFHxA	7.6	ng/L	2.1	0.39	1	06/25/24 11:00	06/27/24 13:57	307-24-4	
PFBA	6.8	ng/L	2.1	0.29	1	06/25/24 11:00	06/27/24 13:57	375-22-4	
PFDS	<0.58	ng/L	2.0	0.58	1	06/25/24 11:00	06/27/24 13:57	335-77-3	
PFHpS	<0.65	ng/L	2.0	0.65	1	06/25/24 11:00	06/27/24 13:57	375-92-8	
PFNS	<0.49	ng/L	2.0	0.49	1	06/25/24 11:00	06/27/24 13:57	68259-12-1	
PFOSA	<0.41	ng/L	2.1	0.41	1	06/25/24 11:00	06/27/24 13:57	754-91-6	
PFPeA	2.2	ng/L	2.1	0.19	1	06/25/24 11:00	06/27/24 13:57	2706-90-3	
PFPeS	<0.26	ng/L	1.9	0.26	1	06/25/24 11:00	06/27/24 13:57	2706-91-4	
PFDoA	<0.44	ng/L	2.1	0.44	1	06/25/24 11:00	06/27/24 13:57	307-55-1	
PFHpA	0.51J	ng/L	2.1	0.24	1	06/25/24 11:00	06/27/24 13:57	375-85-9	
PFHxS	0.35J	ng/L	1.9	0.24	1	06/25/24 11:00	06/27/24 13:57	355-46-4	
PFNA	0.27J	ng/L	2.1	0.21	1	06/25/24 11:00	06/27/24 13:57	375-95-1	
PFOS	1.1J	ng/L	1.9	0.52	1	06/25/24 11:00	06/27/24 13:57	1763-23-1	
PFOA	1.7J	ng/L	2.1	0.27	1	06/25/24 11:00	06/27/24 13:57	335-67-1	
PFTeDA	<0.37	ng/L	2.1	0.37	1	06/25/24 11:00	06/27/24 13:57	376-06-7	
PFTrDA	<0.29	ng/L	2.1	0.29	1	06/25/24 11:00	06/27/24 13:57	72629-94-8	
PFUnA	<0.66	ng/L	2.1	0.66	1	06/25/24 11:00	06/27/24 13:57	2058-94-8	
Surrogates									
13C26:2FTS (S)	363	%	50-150		1	06/25/24 11:00	06/27/24 13:57		S0
13C28:2FTS (S)	246	%	50-150		1	06/25/24 11:00	06/27/24 13:57		S0
13C24:2FTS (S)	222	%	50-150		1	06/25/24 11:00	06/27/24 13:57		S0
13C2-PFDoA (S)	62	%	50-150		1	06/25/24 11:00	06/27/24 13:57		
13C2PFHxDA (S)	40	%	50-150		1	06/25/24 11:00	06/27/24 13:57		S0
13C2-PFTA (S)	53	%	50-150		1	06/25/24 11:00	06/27/24 13:57		
13C3HFPO-DA (S)	46	%	50-150		1	06/25/24 11:00	06/27/24 13:57		S0
13C3-PFBS (S)	66	%	50-150		1	06/25/24 11:00	06/27/24 13:57	375-73-5	
13C3-PFHxS (S)	69	%	50-150		1	06/25/24 11:00	06/27/24 13:57	355-46-4	
13C4-PFBA (S)	45	%	50-150		1	06/25/24 11:00	06/27/24 13:57	375-22-4	S0
13C4-PFHpA (S)	72	%	50-150		1	06/25/24 11:00	06/27/24 13:57	375-85-9	
13C5-PFHxA (S)	75	%	50-150		1	06/25/24 11:00	06/27/24 13:57	307-24-4	
13C5-PFPeA (S)	53	%	50-150		1	06/25/24 11:00	06/27/24 13:57	2706-90-3	
13C6-PFDA (S)	85	%	50-150		1	06/25/24 11:00	06/27/24 13:57	335-76-2	
13C7-PFUDa (S)	79	%	50-150		1	06/25/24 11:00	06/27/24 13:57	2058-94-8	
13C8-PFOA (S)	68	%	50-150		1	06/25/24 11:00	06/27/24 13:57	335-67-1	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50374521

Sample: Effluent Duplicate Lab ID: 50374521001 Collected: 05/29/24 13:27 Received: 05/30/24 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
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PFAS ID NPW

Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178

Pace Analytical Services - Minneapolis

Surrogates

13C8-PFOS (S)	73	%.	50-150		1	06/25/24 11:00	06/27/24 13:57	1763-23-1	
13C8-PFOA (S)	46	%.	50-150		1	06/25/24 11:00	06/27/24 13:57	754-91-6	S0
13C9-PFNA (S)	80	%.	50-150		1	06/25/24 11:00	06/27/24 13:57	375-95-1	
d3-MeFOSAA (S)	78	%.	50-150		1	06/25/24 11:00	06/27/24 13:57	2355-31-9	
d3-NMeFOSA (S)	16	%.	10-150		1	06/25/24 11:00	06/27/24 13:57	31506-32-8	
d5-EtFOSAA (S)	73	%.	50-150		1	06/25/24 11:00	06/27/24 13:57	2991-50-6	
d5-NEtFOSA (S)	14	%.	10-150		1	06/25/24 11:00	06/27/24 13:57	4151-50-2	
d7-NMeFOSE (S)	46	%.	10-150		1	06/25/24 11:00	06/27/24 13:57	24448-09-7	
d9-NEtFOSE (S)	43	%.	10-150		1	06/25/24 11:00	06/27/24 13:57	1691-99-2	

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50374521

Sample: Effluent Lab ID: 50374521002 Collected: 05/29/24 13:25 Received: 05/30/24 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
PFAS ID NPW									
Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178									
Pace Analytical Services - Minneapolis									
11CI-PF3OUdS	<0.40	ng/L	1.9	0.40	1	06/25/24 11:00	06/27/24 13:50	763051-92-9	
4:2 FTS	<0.36	ng/L	1.8	0.36	1	06/25/24 11:00	06/27/24 13:50	757124-72-4	
6:2 FTS	<0.56	ng/L	1.9	0.56	1	06/25/24 11:00	06/27/24 13:50	27619-97-2	
8:2 FTS	<0.78	ng/L	1.9	0.78	1	06/25/24 11:00	06/27/24 13:50	39108-34-4	
9CI-PF3ONS	<0.35	ng/L	1.8	0.35	1	06/25/24 11:00	06/27/24 13:50	756426-58-1	
ADONA	0.46J	ng/L	1.9	0.32	1	06/25/24 11:00	06/27/24 13:50	919005-14-4	
HFPO-DA	<0.25	ng/L	2.0	0.25	1	06/25/24 11:00	06/27/24 13:50	13252-13-6	
NEtFOSAA	<0.56	ng/L	2.0	0.56	1	06/25/24 11:00	06/27/24 13:50	2991-50-6	
NMeFOSAA	<0.77	ng/L	2.0	0.77	1	06/25/24 11:00	06/27/24 13:50	2355-31-9	
PFBS	25.3	ng/L	1.7	0.20	1	06/25/24 11:00	06/27/24 13:50	375-73-5	
PFDA	0.35J	ng/L	2.0	0.25	1	06/25/24 11:00	06/27/24 13:50	335-76-2	
PFHxA	7.4	ng/L	2.0	0.37	1	06/25/24 11:00	06/27/24 13:50	307-24-4	
PFBA	7.5	ng/L	2.0	0.28	1	06/25/24 11:00	06/27/24 13:50	375-22-4	
PFDS	<0.56	ng/L	1.9	0.56	1	06/25/24 11:00	06/27/24 13:50	335-77-3	
PFHpS	<0.62	ng/L	1.9	0.62	1	06/25/24 11:00	06/27/24 13:50	375-92-8	
PFNS	<0.47	ng/L	1.9	0.47	1	06/25/24 11:00	06/27/24 13:50	68259-12-1	
PFOSA	<0.39	ng/L	2.0	0.39	1	06/25/24 11:00	06/27/24 13:50	754-91-6	
PFPeA	1.5J	ng/L	2.0	0.18	1	06/25/24 11:00	06/27/24 13:50	2706-90-3	
PFPeS	<0.25	ng/L	1.9	0.25	1	06/25/24 11:00	06/27/24 13:50	2706-91-4	
PFDoA	<0.43	ng/L	2.0	0.43	1	06/25/24 11:00	06/27/24 13:50	307-55-1	
PFHpA	0.65J	ng/L	2.0	0.23	1	06/25/24 11:00	06/27/24 13:50	375-85-9	
PFHxS	0.66J	ng/L	1.8	0.23	1	06/25/24 11:00	06/27/24 13:50	355-46-4	
PFNA	<0.21	ng/L	2.0	0.21	1	06/25/24 11:00	06/27/24 13:50	375-95-1	
PFOS	0.99J	ng/L	1.8	0.50	1	06/25/24 11:00	06/27/24 13:50	1763-23-1	
PFOA	1.7J	ng/L	2.0	0.26	1	06/25/24 11:00	06/27/24 13:50	335-67-1	
PFTeDA	<0.36	ng/L	2.0	0.36	1	06/25/24 11:00	06/27/24 13:50	376-06-7	
PFTrDA	<0.28	ng/L	2.0	0.28	1	06/25/24 11:00	06/27/24 13:50	72629-94-8	
PFUnA	<0.63	ng/L	2.0	0.63	1	06/25/24 11:00	06/27/24 13:50	2058-94-8	
Surrogates									
13C26:2FTS (S)	351	%	50-150		1	06/25/24 11:00	06/27/24 13:50		S0
13C28:2FTS (S)	243	%	50-150		1	06/25/24 11:00	06/27/24 13:50		S0
13C24:2FTS (S)	230	%	50-150		1	06/25/24 11:00	06/27/24 13:50		S0
13C2-PFDoA (S)	65	%	50-150		1	06/25/24 11:00	06/27/24 13:50		
13C2PFHxDA (S)	38	%	50-150		1	06/25/24 11:00	06/27/24 13:50		S0
13C2-PFTA (S)	51	%	50-150		1	06/25/24 11:00	06/27/24 13:50		
13C3HFPO-DA (S)	46	%	50-150		1	06/25/24 11:00	06/27/24 13:50		S0
13C3-PFBS (S)	64	%	50-150		1	06/25/24 11:00	06/27/24 13:50	375-73-5	
13C3-PFHxS (S)	67	%	50-150		1	06/25/24 11:00	06/27/24 13:50	355-46-4	
13C4-PFBA (S)	45	%	50-150		1	06/25/24 11:00	06/27/24 13:50	375-22-4	S0
13C4-PFHpA (S)	73	%	50-150		1	06/25/24 11:00	06/27/24 13:50	375-85-9	
13C5-PFHxA (S)	76	%	50-150		1	06/25/24 11:00	06/27/24 13:50	307-24-4	
13C5-PFPeA (S)	53	%	50-150		1	06/25/24 11:00	06/27/24 13:50	2706-90-3	
13C6-PFDA (S)	86	%	50-150		1	06/25/24 11:00	06/27/24 13:50	335-76-2	
13C7-PFUdA (S)	80	%	50-150		1	06/25/24 11:00	06/27/24 13:50	2058-94-8	
13C8-PFOA (S)	67	%	50-150		1	06/25/24 11:00	06/27/24 13:50	335-67-1	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50374521

Sample: Effluent Lab ID: 50374521002 Collected: 05/29/24 13:25 Received: 05/30/24 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
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PFAS ID NPW

Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178

Pace Analytical Services - Minneapolis

Surrogates

13C8-PFOS (S)	71	%.	50-150		1	06/25/24 11:00	06/27/24 13:50	1763-23-1	
13C8-PFOA (S)	48	%.	50-150		1	06/25/24 11:00	06/27/24 13:50	754-91-6	S0
13C9-PFNA (S)	75	%.	50-150		1	06/25/24 11:00	06/27/24 13:50	375-95-1	
d3-MeFOSAA (S)	73	%.	50-150		1	06/25/24 11:00	06/27/24 13:50	2355-31-9	
d3-NMeFOSA (S)	20	%.	10-150		1	06/25/24 11:00	06/27/24 13:50	31506-32-8	
d5-EtFOSAA (S)	69	%.	50-150		1	06/25/24 11:00	06/27/24 13:50	2991-50-6	
d5-NEtFOSA (S)	17	%.	10-150		1	06/25/24 11:00	06/27/24 13:50	4151-50-2	
d7-NMeFOSE (S)	53	%.	10-150		1	06/25/24 11:00	06/27/24 13:50	24448-09-7	
d9-NEtFOSE (S)	45	%.	10-150		1	06/25/24 11:00	06/27/24 13:50	1691-99-2	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50374521

Sample: Field Blank Lab ID: 50374521003 Collected: 05/29/24 13:23 Received: 05/30/24 10:15 Matrix: Water									
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
PFAS ID NPW Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178 Pace Analytical Services - Minneapolis									
11CI-PF3OUdS	<0.39	ng/L	1.8	0.39	1	06/25/24 11:00	06/27/24 13:42	763051-92-9	
4:2 FTS	<0.35	ng/L	1.8	0.35	1	06/25/24 11:00	06/27/24 13:42	757124-72-4	
6:2 FTS	<0.55	ng/L	1.8	0.55	1	06/25/24 11:00	06/27/24 13:42	27619-97-2	
8:2 FTS	<0.76	ng/L	1.9	0.76	1	06/25/24 11:00	06/27/24 13:42	39108-34-4	
9CI-PF3ONS	<0.34	ng/L	1.8	0.34	1	06/25/24 11:00	06/27/24 13:42	756426-58-1	
ADONA	<0.31	ng/L	1.8	0.31	1	06/25/24 11:00	06/27/24 13:42	919005-14-4	
HFPO-DA	<0.24	ng/L	1.9	0.24	1	06/25/24 11:00	06/27/24 13:42	13252-13-6	
NEtFOSAA	<0.55	ng/L	1.9	0.55	1	06/25/24 11:00	06/27/24 13:42	2991-50-6	
NMeFOSAA	<0.75	ng/L	1.9	0.75	1	06/25/24 11:00	06/27/24 13:42	2355-31-9	
PFBS	<0.19	ng/L	1.7	0.19	1	06/25/24 11:00	06/27/24 13:42	375-73-5	
PFDA	<0.24	ng/L	1.9	0.24	1	06/25/24 11:00	06/27/24 13:42	335-76-2	
PFHxA	<0.36	ng/L	1.9	0.36	1	06/25/24 11:00	06/27/24 13:42	307-24-4	
PFBA	<0.27	ng/L	1.9	0.27	1	06/25/24 11:00	06/27/24 13:42	375-22-4	
PFDS	<0.55	ng/L	1.9	0.55	1	06/25/24 11:00	06/27/24 13:42	335-77-3	
PFHpS	<0.60	ng/L	1.8	0.60	1	06/25/24 11:00	06/27/24 13:42	375-92-8	
PFNS	<0.45	ng/L	1.8	0.45	1	06/25/24 11:00	06/27/24 13:42	68259-12-1	
PFOSA	<0.38	ng/L	1.9	0.38	1	06/25/24 11:00	06/27/24 13:42	754-91-6	
PFPeA	<0.18	ng/L	1.9	0.18	1	06/25/24 11:00	06/27/24 13:42	2706-90-3	
PFPeS	<0.25	ng/L	1.8	0.25	1	06/25/24 11:00	06/27/24 13:42	2706-91-4	
PFDoA	<0.41	ng/L	1.9	0.41	1	06/25/24 11:00	06/27/24 13:42	307-55-1	
PFHpA	<0.23	ng/L	1.9	0.23	1	06/25/24 11:00	06/27/24 13:42	375-85-9	
PFHxS	<0.22	ng/L	1.7	0.22	1	06/25/24 11:00	06/27/24 13:42	355-46-4	
PFNA	<0.20	ng/L	1.9	0.20	1	06/25/24 11:00	06/27/24 13:42	375-95-1	
PFOS	<0.49	ng/L	1.8	0.49	1	06/25/24 11:00	06/27/24 13:42	1763-23-1	
PFOA	<0.26	ng/L	1.9	0.26	1	06/25/24 11:00	06/27/24 13:42	335-67-1	
PFTeDA	<0.35	ng/L	1.9	0.35	1	06/25/24 11:00	06/27/24 13:42	376-06-7	
PFTrDA	<0.27	ng/L	1.9	0.27	1	06/25/24 11:00	06/27/24 13:42	72629-94-8	
PFUnA	<0.61	ng/L	1.9	0.61	1	06/25/24 11:00	06/27/24 13:42	2058-94-8	
Surrogates									
13C26:2FTS (S)	74	%	50-150		1	06/25/24 11:00	06/27/24 13:42		
13C28:2FTS (S)	84	%	50-150		1	06/25/24 11:00	06/27/24 13:42		
13C24:2FTS (S)	72	%	50-150		1	06/25/24 11:00	06/27/24 13:42		
13C2-PFDoA (S)	58	%	50-150		1	06/25/24 11:00	06/27/24 13:42		
13C2PFHxDA (S)	39	%	50-150		1	06/25/24 11:00	06/27/24 13:42		S0
13C2-PFTA (S)	52	%	50-150		1	06/25/24 11:00	06/27/24 13:42		
13C3HFPO-DA (S)	76	%	50-150		1	06/25/24 11:00	06/27/24 13:42		
13C3-PFBS (S)	82	%	50-150		1	06/25/24 11:00	06/27/24 13:42	375-73-5	
13C3-PFHxS (S)	83	%	50-150		1	06/25/24 11:00	06/27/24 13:42	355-46-4	
13C4-PFBA (S)	79	%	50-150		1	06/25/24 11:00	06/27/24 13:42	375-22-4	
13C4-PFHpA (S)	84	%	50-150		1	06/25/24 11:00	06/27/24 13:42	375-85-9	
13C5-PFHxA (S)	81	%	50-150		1	06/25/24 11:00	06/27/24 13:42	307-24-4	
13C5-PFPeA (S)	80	%	50-150		1	06/25/24 11:00	06/27/24 13:42	2706-90-3	
13C6-PFDA (S)	73	%	50-150		1	06/25/24 11:00	06/27/24 13:42	335-76-2	
13C7-PFUDa (S)	64	%	50-150		1	06/25/24 11:00	06/27/24 13:42	2058-94-8	
13C8-PFOA (S)	85	%	50-150		1	06/25/24 11:00	06/27/24 13:42	335-67-1	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50374521

Sample: Field Blank Lab ID: 50374521003 Collected: 05/29/24 13:23 Received: 05/30/24 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
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PFAS ID NPW

Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178

Pace Analytical Services - Minneapolis

Surrogates

13C8-PFOS (S)	71	%.	50-150		1	06/25/24 11:00	06/27/24 13:42	1763-23-1	
13C8-PFOA (S)	68	%.	50-150		1	06/25/24 11:00	06/27/24 13:42	754-91-6	
13C9-PFNA (S)	80	%.	50-150		1	06/25/24 11:00	06/27/24 13:42	375-95-1	
d3-MeFOSAA (S)	57	%.	50-150		1	06/25/24 11:00	06/27/24 13:42	2355-31-9	
d3-NMeFOSA (S)	12	%.	10-150		1	06/25/24 11:00	06/27/24 13:42	31506-32-8	
d5-EtFOSAA (S)	59	%.	50-150		1	06/25/24 11:00	06/27/24 13:42	2991-50-6	
d5-NEtFOSA (S)	9	%.	10-150		1	06/25/24 11:00	06/27/24 13:42	4151-50-2	S0
d7-NMeFOSE (S)	54	%.	10-150		1	06/25/24 11:00	06/27/24 13:42	24448-09-7	
d9-NEtFOSE (S)	50	%.	10-150		1	06/25/24 11:00	06/27/24 13:42	1691-99-2	

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50374521

QC Batch: 952249

Analysis Method: ENV-SOP-MIN4-0178

QC Batch Method: ENV-SOP-MIN4-0178

Analysis Description: PFAS Compounds, Water

Laboratory:

Pace Analytical Services - Minneapolis

Associated Lab Samples: 50374521001, 50374521002, 50374521003

METHOD BLANK: 4979336

Matrix: Water

Associated Lab Samples: 50374521001, 50374521002, 50374521003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11CI-PF3OUdS	ng/L	<0.39	1.8	0.39	06/27/24 11:25	
4:2 FTS	ng/L	<0.35	1.8	0.35	06/27/24 11:25	
6:2 FTS	ng/L	<0.55	1.8	0.55	06/27/24 11:25	
8:2 FTS	ng/L	<0.77	1.9	0.77	06/27/24 11:25	
9CI-PF3ONS	ng/L	<0.34	1.8	0.34	06/27/24 11:25	
ADONA	ng/L	<0.31	1.8	0.31	06/27/24 11:25	
HFPO-DA	ng/L	<0.24	1.9	0.24	06/27/24 11:25	
NETFOSAA	ng/L	<0.55	1.9	0.55	06/27/24 11:25	
NMeFOSAA	ng/L	<0.76	1.9	0.76	06/27/24 11:25	
PFBA	ng/L	<0.27	1.9	0.27	06/27/24 11:25	
PFBS	ng/L	<0.20	1.7	0.20	06/27/24 11:25	
PFDA	ng/L	<0.24	1.9	0.24	06/27/24 11:25	
PFDoA	ng/L	<0.42	1.9	0.42	06/27/24 11:25	
PFDS	ng/L	<0.55	1.9	0.55	06/27/24 11:25	
PFHpA	ng/L	<0.23	1.9	0.23	06/27/24 11:25	
PFHpS	ng/L	<0.61	1.8	0.61	06/27/24 11:25	
PFHxA	ng/L	<0.37	1.9	0.37	06/27/24 11:25	
PFHxS	ng/L	<0.23	1.8	0.23	06/27/24 11:25	
PFNA	ng/L	<0.20	1.9	0.20	06/27/24 11:25	
PFNS	ng/L	<0.46	1.9	0.46	06/27/24 11:25	
PFOA	ng/L	<0.26	1.9	0.26	06/27/24 11:25	
PFOS	ng/L	<0.49	1.8	0.49	06/27/24 11:25	
PFOSA	ng/L	<0.39	1.9	0.39	06/27/24 11:25	
PFPeA	ng/L	<0.18	1.9	0.18	06/27/24 11:25	
PFPeS	ng/L	<0.25	1.8	0.25	06/27/24 11:25	
PFTeDA	ng/L	<0.35	1.9	0.35	06/27/24 11:25	
PFTTrDA	ng/L	<0.27	1.9	0.27	06/27/24 11:25	
PFUnA	ng/L	<0.62	1.9	0.62	06/27/24 11:25	
13C2-PFDoA (S)	%	67	50-150		06/27/24 11:25	
13C2-PFTA (S)	%	68	50-150		06/27/24 11:25	
13C24:2FTS (S)	%	100	50-150		06/27/24 11:25	
13C26:2FTS (S)	%	103	50-150		06/27/24 11:25	
13C28:2FTS (S)	%	83	50-150		06/27/24 11:25	
13C2PFHxDA (S)	%	44	50-150		06/27/24 11:25	S0
13C3-PFBS (S)	%	83	50-150		06/27/24 11:25	
13C3-PFHxS (S)	%	83	50-150		06/27/24 11:25	
13C3HFPO-DA (S)	%	77	50-150		06/27/24 11:25	
13C4-PFBA (S)	%	79	50-150		06/27/24 11:25	
13C4-PFHpA (S)	%	85	50-150		06/27/24 11:25	
13C5-PFHxA (S)	%	83	50-150		06/27/24 11:25	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50374521

METHOD BLANK: 4979336

Matrix: Water

Associated Lab Samples: 50374521001, 50374521002, 50374521003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
13C5-PFPeA (S)	%	83	50-150		06/27/24 11:25	
13C6-PFDA (S)	%	77	50-150		06/27/24 11:25	
13C7-PFUDa (S)	%	70	50-150		06/27/24 11:25	
13C8-PFOA (S)	%	87	50-150		06/27/24 11:25	
13C8-PFOS (S)	%	78	50-150		06/27/24 11:25	
13C8-PFOSA (S)	%	71	50-150		06/27/24 11:25	
13C9-PFNA (S)	%	84	50-150		06/27/24 11:25	
d3-MeFOSAA (S)	%	71	50-150		06/27/24 11:25	
d3-NMeFOSA (S)	%	15	10-150		06/27/24 11:25	
d5-EtFOSAA (S)	%	69	50-150		06/27/24 11:25	
d5-NEtFOSA (S)	%	13	10-150		06/27/24 11:25	
d7-NMeFOSE (S)	%	63	10-150		06/27/24 11:25	
d9-NEtFOSE (S)	%	56	10-150		06/27/24 11:25	

LABORATORY CONTROL SAMPLE & LCSD: 4979337

4979338

Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
11CI-PF3OUdS	ng/L	3.7	2.6	2.6	71	71	70-140	1	30	
4:2 FTS	ng/L	3.6	3.1	3.1	84	87	63-143	1	30	
6:2 FTS	ng/L	3.7	3.3	3.0	90	84	64-140	9	30	
8:2 FTS	ng/L	3.7	3.2	3.4	86	92	67-138	5	30	
9CI-PF3ONS	ng/L	3.6	3.0	2.9	81	81	70-140	3	30	
ADONA	ng/L	3.7	3.1	3.1	86	88	70-140	0	30	
HFPO-DA	ng/L	3.9	3.4	3.3	86	86	70-140	3	30	
NEtFOSAA	ng/L	3.9	2.8	3.3	73	87	61-135	15	30	
NMeFOSAA	ng/L	3.9	3.4	3.2	88	85	65-136	5	30	
PFBA	ng/L	3.9	3.0	3.2	78	85	73-129	6	30	
PFBS	ng/L	3.4	2.9	2.9	84	85	72-130	0	30	
PFDA	ng/L	3.9	3.4	3.3	88	87	71-129	3	30	
PFDoA	ng/L	3.9	3.3	3.3	86	87	72-134	1	30	
PFDS	ng/L	3.7	2.9	2.3	77	62	53-142	24	30	
PFHpA	ng/L	3.9	3.2	3.3	81	86	72-130	4	30	
PFHpS	ng/L	3.7	3.3	3.3	89	91	69-134	0	30	
PFHxA	ng/L	3.9	3.2	3.3	82	87	72-129	4	30	
PFHxS	ng/L	3.6	3.1	3.1	86	88	68-131	0	30	
PFNA	ng/L	3.9	3.2	3.2	82	85	69-130	1	30	
PFNS	ng/L	3.7	2.8	3.1	76	84	69-127	9	30	
PFOA	ng/L	3.9	3.4	3.4	88	90	71-133	0	30	
PFOS	ng/L	3.6	2.8	2.8	77	80	65-140	2	30	
PFOSA	ng/L	3.9	3.2	3.1	83	82	67-137	4	30	
PFPeA	ng/L	3.9	3.4	3.3	88	87	72-129	3	30	
PFPeS	ng/L	3.6	2.9	3.0	80	84	71-127	2	30	
PFTeDA	ng/L	3.9	3.3	3.2	84	84	71-132	2	30	
PFTTrDA	ng/L	3.9	3.4	3.7	87	96	65-144	8	30	

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50374521

LABORATORY CONTROL SAMPLE & LCSD: 4979337										
4979338										
Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
PFUnA	ng/L	3.9	3.1	3.4	80	90	69-133	9	30	
13C2-PFDoA (S)	%.				76	74	50-150			
13C2-PFTA (S)	%.				77	75	50-150			
13C24:2FTS (S)	%.				103	104	50-150			
13C26:2FTS (S)	%.				90	112	50-150			
13C28:2FTS (S)	%.				85	87	50-150			
13C2PFHxDA (S)	%.				56	59	50-150			
13C3-PFBS (S)	%.				84	86	50-150			
13C3-PFHxS (S)	%.				84	88	50-150			
13C3HFPO-DA (S)	%.				84	83	50-150			
13C4-PFBA (S)	%.				80	84	50-150			
13C4-PFHpA (S)	%.				88	90	50-150			
13C5-PFHxA (S)	%.				85	87	50-150			
13C5-PFPeA (S)	%.				81	87	50-150			
13C6-PFDA (S)	%.				82	87	50-150			
13C7-PFUdA (S)	%.				79	81	50-150			
13C8-PFOA (S)	%.				88	92	50-150			
13C8-PFOS (S)	%.				81	85	50-150			
13C8-PFOSA (S)	%.				79	78	50-150			
13C9-PFNA (S)	%.				87	90	50-150			
d3-MeFOSAA (S)	%.				68	75	50-150			
d3-NMeFOSA (S)	%.				36	26	10-150			
d5-EtFOSAA (S)	%.				75	70	50-150			
d5-NEtFOSA (S)	%.				37	20	10-150			
d7-NMeFOSE (S)	%.				71	69	10-150			
d9-NEtFOSE (S)	%.				66	63	10-150			

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QUALIFIERS

Project: Effluent PFAS
Pace Project No.: 50374521

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

S0 Surrogate recovery outside laboratory control limits.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Effluent PFAS
Pace Project No.: 50374521

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50374521001	Effluent Duplicate	ENV-SOP-MIN4-0178	952249	ENV-SOP-MIN4-0178	953832
50374521002	Effluent	ENV-SOP-MIN4-0178	952249	ENV-SOP-MIN4-0178	953832
50374521003	Field Blank	ENV-SOP-MIN4-0178	952249	ENV-SOP-MIN4-0178	953832

REPORT OF LABORATORY ANALYSIS



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>5-30-24</u>	Evaluated By: <u>WDC</u>	WO#: 50374521	
Client: <u>City of Cadillac</u>	PM: <u>BJH</u>	PM: BJH	Due Date: 06/28/24
Lab Notified of Rush or Short Holds: <u>YES</u> <u>NO</u> <u>NA</u>		CLIENT: GR-Cadillac	
Project Received Via: <u>FedEx</u> <u>UPS</u> <u>Client</u> <u>Pace Courier</u> Other: _____			Comments:
Custody Seal Present and Intact:	<u>YES</u>	<u>NO</u>	<u>NA</u>
Received Sample Information Form (SIF): Drinking Waters Only	<u>YES</u>	<u>NO</u>	<u>NA</u>
Short Hold Present (≤ 48 Hours):	<u>YES</u>	<u>NO</u>	
Sample Received in Hold:	<u>YES</u>	<u>NO</u>	
Custody Signature Present:	<u>YES</u>	<u>NO</u>	
Collector Signature Present:	<u>YES</u>	<u>NO</u>	
Sample Collected Today and On Ice:	<u>YES</u>	<u>NO</u>	<u>N/A</u>
IR Gun #: <u>350</u> <u>351</u> Therm #: <u>282</u> <u>283</u>	Temp. should be 0°C - 6°C (Initial/Corrected)		
Ice Type: <u>WET Bagged / WET Loose</u> <u>BLUE</u> <u>NONE</u>	1. Cooler Temp. Upon Receipt: <u>15/11</u> °C		
Ice Location: <u>TOP</u> <u>BOTTOM</u> <u>MIDDLE</u> <u>DISPERSED</u>	2. Cooler Temp. Upon Receipt: _____ °C		
Temp Blank Received:	<u>YES</u>	<u>NO</u>	
Sample Label Matches COC (ID/Date/Time):	<u>YES</u>	<u>NO</u>	
Container Intact:	<u>YES</u>	<u>NO</u>	
Correct Container:	<u>YES</u>	<u>NO</u>	
Sufficient Volume:	<u>YES</u>	<u>NO</u>	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl	<u>YES</u>	<u>NO</u>	<u>N/A</u>
Residual Chlorine Absent: <u>Cl₂ Strip Lot #: _____</u> Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide	<u>YES</u>	<u>NO</u>	<u>N/A</u>
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	<u>YES</u>	<u>NO</u>	<u>N/A</u>
Trip Blank Received: <u>HCl</u> <u>MeOH</u> Other: _____	<u>YES</u>	<u>NO</u>	<u>ON HOLD</u>
Comments:	3. Cooler Temp. Upon Receipt: _____ °C		
	4. Cooler Temp. Upon Receipt: _____ °C		
	Non-Conformance Form Required: <u>YES</u> <u>NO</u>		